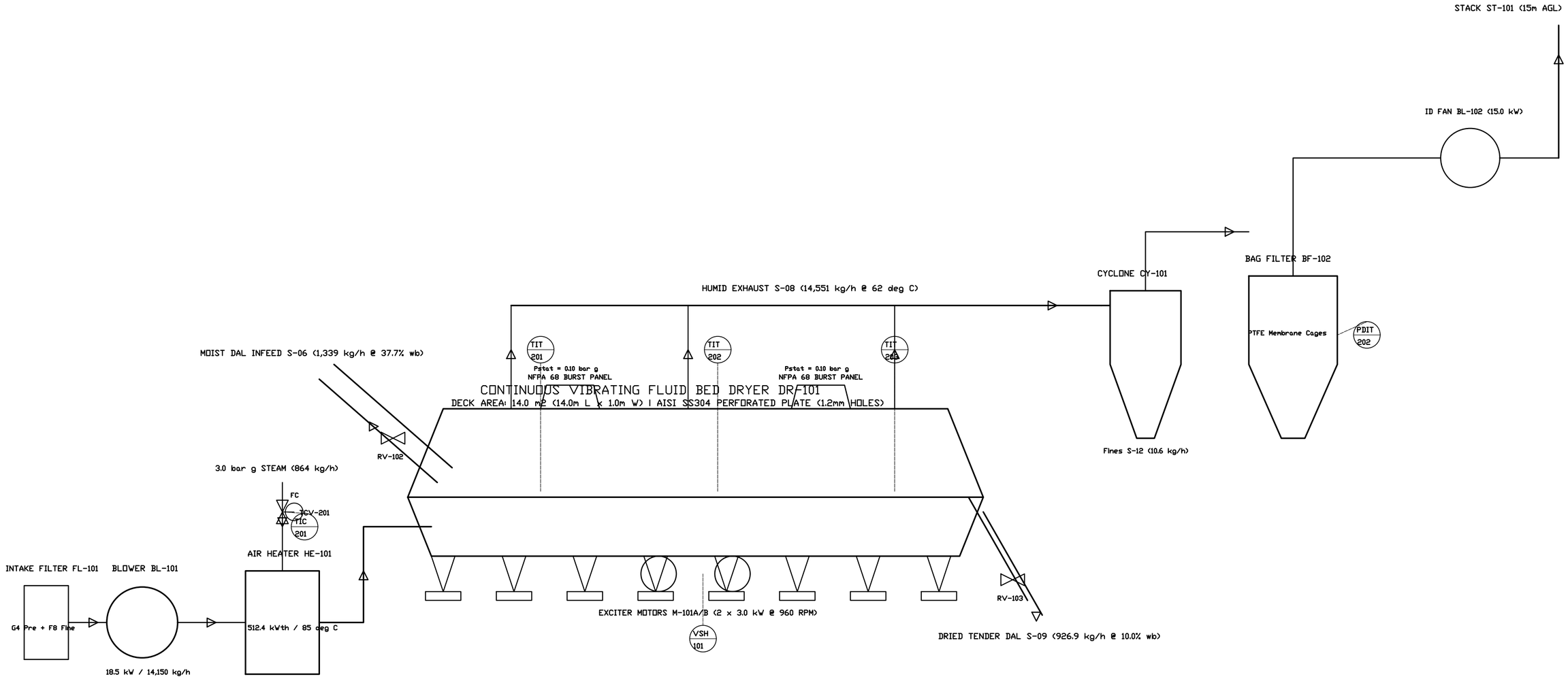


DRYER CONTROL PHILOSOPHY & SAFETY INTERLOCK MATRIX (ISA-84 / SIL-1):

1. TIC-201: Modulates 3" steam control valve TCV-201 to maintain fluidizing air at 85.0 deg C +/- 1.0 deg C.

2. I-02 Interlock: Low airflow (FIT-201 < 10,000 kg/h) or Blower BL-101 trip immediately failsafe-closes TCV-201.

3. I-03 Interlock: High-High bed temperature (TIT-202 > 95 deg C) trips infeed rotary valve RV-102 and alarms operator.

4. PDIT-202: Differential pressure across baghouse triggers sequential reverse pulse-jet cleaning @ 5.0 bar g air.

5. Vibration Interlock: High vibration switch VSH-101 trips exciter motors M-101A/B upon mechanical resonance.

6. Explosion Protection: Certified NFPA 68 burst panels vent deflagration safely upward to outdoor atmosphere.

EQUIPMENT SPECIFICATIONS & DESIGN PARAMETERS (DR-101 TRAIN):

DR-101: 14.0m L x 1.0m W x 2.8m H | Deck Area: 14.0 m2 | MOC: AISI SS304 contact, MS frame | 6,850 kg.

HE-101: 512.4 kWth finned-tube steam battery | Steam: 864 kg/h @ 3.0 bar g | Area: 42.0 m2 | 1,200 kg.

BL-101: Centrifugal backward-curved fan | 14,150 kg/h (11,800 m3/h) @ 2,800 Pa SP | 18.5 kW TEFC IP55.

CY-101: High-efficiency cyclone separator | 4.0m H x 1.2m Dia | Fines cut size d50 = 8 um | 850 kg.

BF-102: Pulse-jet baghouse | 64 PTFE needle-felt bags | Air-to-Cloth: 1.1 m/min | Residual dust < 5 mg/Nm3.

BL-102: Induced draft exhaust fan | 14,550 kg/h @ 2,200 Pa SP | 15.0 kW TEFC IP55 motor.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - FLUID BED DRYER & AIR SYSTEM

DISCIPLINE: PIPING | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-005	REV: V8.0	SHEET: 5 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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